



TOOLS FOR TROUBLE

SHOOTER THEORY

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

1. *Computer won't power on: Measure DC voltage at the power-supply outputs to determine whether the PSU is delivering the expected voltage.*
2. *Laptop battery not charging: Test the battery's voltage and check continuity through the charging port to identify a weak battery or broken connection.*
3. *Smartphone overheating or shutting down: Use resistance or diode mode with the device powered off to look for a short circuit between power and ground on the motherboard.*

Always disconnect power before continuity or resistance tests; voltage measurements on powered devices require correct meter settings and careful probing

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video. Focus on the codes shown and how they relate to hardware issues.

The POST card (a PCI/ISA diagnostic card) displays two-digit hexadecimal codes on dual 7-segment LEDs (current and previous), plus power-rail/reset LEDs, reporting BIOS Power-On Self-Test progress via I/O port 80h.

In demos of testing faulty motherboards (e.g., socket AM2 boards), it typically advances through codes then sticks (commonly at 05, d5, or similar).

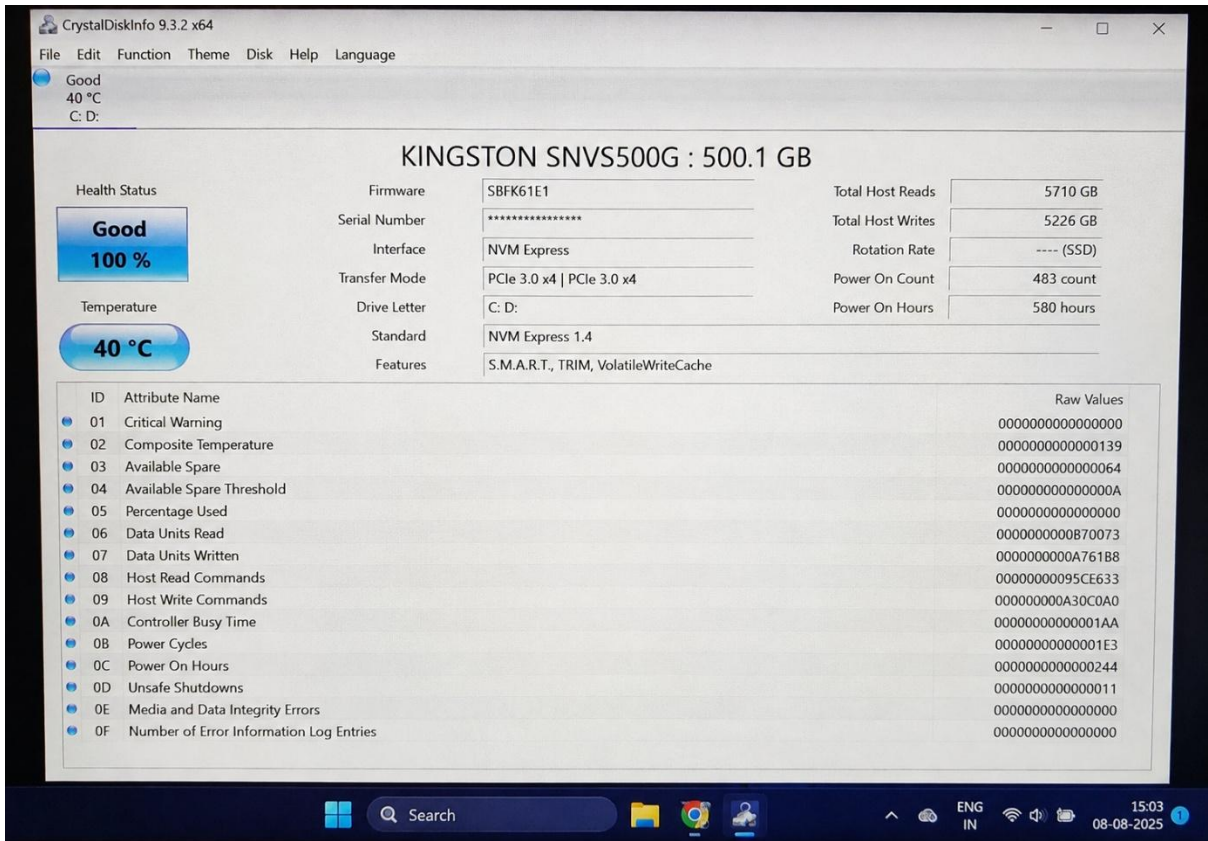
This stuck code points to a hardware failure—often the chipset (needing reflow/heating), incompatible/missing RAM, CPU, or video—allowing isolation of the issue before a full boot or display appears.

Once components are swapped or the chipset addressed, codes progress to 00 (normal completion) and the board can POST successfully.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy,

or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.



4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

<i>Tool</i>	<i>What it checks</i>	<i>Example Problem Detected</i>
<i>Multimeter</i>	<i>Electrical values (voltage, continuity, resistance, current) on power supplies, cables, motherboard traces, and components</i>	<i>No or incorrect voltage from the PSU (e.g., 12 V rail reading 0 V or far below specification), indicating a failed power supply or bad power connector</i>
<i>POST Card</i>	<i>Power-On Self-Test (POST) codes and boot-process status via motherboard expansion slot or diagnostic header</i>	<i>Specific POST error code (e.g., indicating CPU failure, memory initialization error, or video subsystem problem) that prevents the system from completing POST</i>
<i>Software Diagnostics</i>	<i>Hardware functionality, stability, and performance through OS-level or bootable utilities (memory tests, stress tests, sensor monitoring, driver/device checks)</i>	<i>Failing RAM modules or bad memory addresses detected by MemTest86 / Windows Memory Diagnostic, or overheating / thermal throttling shown by monitoring software</i>